

CSE 715: Computer Graphics

Chapter 12 — Ray Tracing

Ray vs Vector, Ray Equation, Ray–Sphere/Ray–Plane Intersection,
Point/Plane Side Test

University of Chittagong — 7th Semester B.Sc. (Engg.) Examination

Source: Schaum’s Outline of Computer Graphics (Xiang & Plastock), Chapter 12 — notation follows the book exactly ($\mathbf{r}(t) = \mathbf{s} + t\mathbf{d}$, A, B, C quadratic coefficients)

Yield note (re-verified against all 5 raw scanned papers 2026-07-07): 2020 has zero ray-tracing content of any kind — its Section-B Q7/Q8 are point left/right test, polygon orientation, 3D rotation matrices, projections, and point-obscures-via-viewpoint, none of which touch rays. The previous `PastPapers.md` table wrongly credited 2020 with ray-sphere/ray-vector/ray-equation questions — corrected in this pass. Real spread: ray-sphere (2021, 2022, 2023), ray vs vector (2021, 2022, 2024), ray equation point-finding (2021, 2022 only), plus two new subtopics split out today: ”ray tracing integrates hidden-surface+projection” (2021, 2022-related, 2024) and the 2024-only 3D point/plane same-side test.

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Theoretical Framework

Ray vs Vector

A **vector** is defined only by **direction and magnitude** — it is translation-invariant; the same vector can be drawn starting anywhere. A **ray** is defined by a **direction and a fixed starting point $\mathbf{s}$** , and extends in that one direction only for $t \geq 0$ (a half-line, not a full line). Two rays with identical direction but different starting points are different rays, whereas the corresponding vectors would be considered the same.

Parametric Vector Representation of a Ray

$$\mathbf{r}(t) = \mathbf{s} + t\mathbf{d}, \quad t \geq 0$$

where $\mathbf{s}$ is the starting point and $\mathbf{d}$ the direction vector. Substituting a value of t gives the coordinates of the point on the ray at that parameter value (applied component-wise to $\mathbf{I}, \mathbf{J}, \mathbf{K}$).

The Recursive Ray-Tracing Algorithm

Each pixel's color comes from a **primary ray** traced from the viewpoint through the pixel into the scene. At the closest intersection point, the color is the sum of three contributions, each computed via a **secondary ray**:

- **Local contribution** (shadow/illumination ray): direct light from source(s), via the Phong formula; blocked $\Rightarrow$ in shadow.
- **Reflected contribution** (specularly reflected ray): inter-object reflection, found by recursively ray-tracing the mirror-reflection direction.
- **Transmitted contribution** (specularly transmitted/refracted ray): light through a transparent surface, direction given by Snell's law $\sin \alpha / \sin \beta = \eta_2 / \eta_1$.

Full formula: $I = I_a k_a + \sum_i I_{p_i} (k_d \mathbf{L}_i \cdot \mathbf{N} + k_s (\mathbf{R}_i \cdot \mathbf{V})^n) + k_r I_r + k_t I_t$, where I_r, I_t are obtained by recursively treating the reflected/transmitted ray as a new primary ray (up to a depth limit). **Why ray tracing is efficient/integrates hidden-surface removal + projection (Solved Prob. 12.1)**: each primary ray is a *projector* mapping a surface point P to its image P' on the view plane. If all primary rays emanate from one viewpoint, this *is* perspective projection; if all primary rays are parallel, this *is* parallel projection. If a primary ray intersects several object surfaces, only the **closest** one to the view plane determines the pixel's color — every other surface it hits is automatically treated as hidden and discarded. So a single ray/surface-intersection test simultaneously performs projection *and* hidden-surface removal, instead of needing them as separate pipeline stages.

Ray–Sphere Intersection

Sphere: center $\mathbf{c}$, radius R . Substituting $\mathbf{r}(t) = \mathbf{s} + t\mathbf{d}$ into $|\mathbf{p} - \mathbf{c}| = R$ and squaring gives a quadratic in t :

$$At^2 + 2Bt + C = 0, \quad A = |\mathbf{d}|^2, \quad B = (\mathbf{s} - \mathbf{c}) \cdot \mathbf{d}, \quad C = |\mathbf{s} - \mathbf{c}|^2 - R^2$$

$$t = \frac{-B \pm \sqrt{B^2 - AC}}{A}$$

$B^2 - AC < 0$	no intersection
$B^2 - AC = 0$	ray (or its negative extension) touches the sphere
$B^2 - AC > 0$	two t values t_1, t_2 : both $< 0 \Rightarrow$ negative extension only (no real intersection); one $= 0 \Rightarrow$ ray starts on sphere, intersects only if the other root > 0 ; differ in sign $\Rightarrow$ ray starts <i>inside</i> the sphere, intersects once (at the positive root); both positive $\Rightarrow$ ray intersects twice (enters then exits), smaller root = nearer intersection

Ray–Plane Intersection & Point/Plane Same-Side Test

For a plane with normal $\mathbf{n} = x_n\mathbf{I} + y_n\mathbf{J} + z_n\mathbf{K}$ through point P_0 , the plane equation is $x_n(x - x_0) + y_n(y - y_0) + z_n(z - z_0) = 0$, i.e. in the form $ax + by + cz = d$ where $(a, b, c) = \mathbf{n}$. **Same-side test:** define $f(x, y, z) = ax + by + cz - d$; two points are on the **same side** of the plane iff f has the **same sign** at both (this is the direct 3D analogue of the 2D point left/right-of-line test in the Chapter 11 wiki page). **Ray–plane intersection:** substituting $\mathbf{s} + t\mathbf{d}$ for (x, y, z) and solving for t gives $t = \frac{\mathbf{n} \cdot \mathbf{p}}{\mathbf{n} \cdot \mathbf{d}}$ where $\mathbf{p} = P_0 - \mathbf{s}$; $\mathbf{n} \cdot \mathbf{d} = 0 \Rightarrow$ ray parallel to plane (no intersection); $t < 0 \Rightarrow$ only the negative extension intersects; $t > 0 \Rightarrow$ the ray itself intersects.

Examination 2021 — Q8(a), Q8(b), Q8(c) & Q8(e)

Q8(a) — Ray Tracing Integrates Hidden-Surface + Projection [1.5 marks]

Question: Describe how hidden surface removal and projection are integrated into the ray-tracing process.

See §1.3: each primary ray is a projector mapping $P \rightarrow P'$; rays converging at one viewpoint = perspective, parallel rays = parallel projection; when a ray hits multiple surfaces, only the closest to the view plane is kept (the rest are hidden surfaces,

removed automatically).

Q8(b) — Ray Equation Point-Finding [2.5 marks]

Question: A ray is represented by $\mathbf{r}(t) = \mathbf{s} + t\mathbf{d}$ where $\mathbf{s} = 2\mathbf{I} + \mathbf{J} - 3\mathbf{K}$ and $\mathbf{d} = \mathbf{I} + 2\mathbf{K}$. Find the coordinates of the points on the ray that correspond to $t = 0, 1, 7, 4, 3$.

$$\mathbf{r}(t) = (2 + t)\mathbf{I} + \mathbf{J} + (-3 + 2t)\mathbf{K}:$$

$$t=0: (2, 1, -3) \quad t=1: (3, 1, -1) \quad t=7: (9, 1, 11) \quad t=4: (6, 1, 5) \quad t=3: (5, 1, 3)$$

Q8(c) — Ray vs Vector [1 mark]

Question: What is the difference between a vector and a ray?

See §1.1: a vector has direction+magnitude only (translation-invariant); a ray has direction+a fixed starting point and extends one-sidedly ($t \geq 0$).

Q8(e) — Ray–Sphere Intersection (Two Spheres) [3.75 marks]

Question: Let S_1 be a sphere of radius 8 centered at $(2, 4, 1)$ and S_2 a sphere of radius 10 centered at $(10, -2, -5)$. Determine if a ray with $\mathbf{s} = 2\mathbf{J} + 5\mathbf{K}$ and $\mathbf{d} = \mathbf{I} - 2\mathbf{K}$ intersects the spheres.

This is Schaum's own worked example (Solved Prob. 12.15) — identical numbers. $A = |\mathbf{d}|^2 = 1 + 4 = 5$. S_1 : $\mathbf{s} - \mathbf{c}_1 = (-2, -2, 4)$, $B = (-2)(1) + (-2)(0) + (4)(-2) = -10$, $C = 4 + 4 + 16 - 64 = -40$. $B^2 - AC = 100 + 200 = 300 > 0$: $t_1 = 2(1 + \sqrt{3}) \approx 5.464 > 0$, $t_2 = 2(1 - \sqrt{3}) \approx -1.464 < 0$ — signs differ $\Rightarrow$ ray starts **inside** S_1 , exits at $t \approx 5.464$. S_2 : $\mathbf{s} - \mathbf{c}_2 = (-10, 4, 10)$, $B = -10 - 20 = -30$, $C = 100 + 16 + 100 - 100 = 116$. $B^2 - AC = 900 + 580 = 320 > 0$: $t_3 = 6 + 1.6\sqrt{5} \approx 9.578$, $t_4 = 6 - 1.6\sqrt{5} \approx 2.422$ — both positive $\Rightarrow$ ray enters S_2 at $t \approx 2.422$, exits at $t \approx 9.578$. **Conclusion:** comparing $t_2 < 0 < t_4(2.422) < t_1(5.464) < t_3(9.578)$ — the ray starts inside S_1 , **enters S_2 before leaving S_1** , then finally exits S_2 (the two spheres overlap).

Examination 2022 — Q7(c)-related, Q8(a), Q8(c) & Q8(d)

Q7(c)-related — Ray Tracing Benefits [3 marks, bundled with homogeneous coordinates]

Question: What is the purpose of homogeneous coordinates? How does ray tracing technique incorporate the benefits of hidden surface removal and projection techniques?

Homogeneous coordinates part (Related, primary home = Ch4/Ch7 transformation topics): represent a point (x, y, z) as $(x, y, z, 1)$ so that translation — otherwise not expressible as a matrix multiply — becomes a matrix product like rotation/scaling, letting all affine transforms compose into a single matrix; also enables perspective projection via a 4th-row divide. **Ray-tracing part:** identical to §1.3/Q8(a) 2021 — primary rays double as projectors (viewpoint-converging = perspective, parallel = parallel projection) and automatically resolve hidden surfaces by keeping only the nearest intersection.

Q8(a) — Ray vs Vector [1 mark]

Question: Show that a ray is not a vector.

See §1.1: a vector is fully specified by direction+magnitude and is position-free; a ray additionally requires a fixed starting point and only extends in one direction ($t \geq 0$), so two rays with the same direction but different origins are different rays even though the corresponding vectors would be identical — hence a ray carries more information than a vector and is not simply a vector.

Q8(c) — Ray Equation Point-Finding [3 marks]

Question: A ray is represented by $\mathbf{r}(t) = \mathbf{s} + t\mathbf{d}$ where $\mathbf{s} = 2\mathbf{I} - 3\mathbf{J} - 7\mathbf{K}$ and $\mathbf{d} = \mathbf{I} - \mathbf{J} - 2\mathbf{K}$. Find the coordinates of the points on the ray that correspond to $t = 1, 2, -4, 3, 7$.

$$\mathbf{r}(t) = (2 + t)\mathbf{I} + (-3 - t)\mathbf{J} + (-7 - 2t)\mathbf{K}:$$

$$t=1:(3, -4, -9) \quad t=2:(4, -5, -11) \quad t=-4:(-2, 1, 1) \quad t=3:(5, -6, -13) \quad t=7:(9, -10, -21)$$

Q8(d) — Ray–Sphere Intersection [4 marks]

Question: Assume a sphere of radius 2 with center located at the origin. If a ray originates from the point $(3, 0, 0)$ in the direction $(-3, 1, 0)$, determine the intersection

point(s) of that ray with the sphere.

$\mathbf{s} = (3, 0, 0)$, $\mathbf{d} = (-3, 1, 0)$, $\mathbf{c} = (0, 0, 0)$, $R = 2$. $A = |\mathbf{d}|^2 = 9 + 1 = 10$. $B = (\mathbf{s} - \mathbf{c}) \cdot \mathbf{d} = (3)(-3) + 0 + 0 = -9$. $C = |\mathbf{s} - \mathbf{c}|^2 - R^2 = 9 - 4 = 5$. $B^2 - AC = 81 - 50 = 31 > 0$:

$$t = \frac{9 \pm \sqrt{31}}{10} \Rightarrow t_1 \approx 1.457, t_2 \approx 0.343$$

Both positive $\Rightarrow$ ray intersects the sphere **twice**. Near point ($t_2 \approx 0.343$): $(3 - 3(0.343), 0.343, 0) \approx (1.970, 0.343, 0)$. Far point ($t_1 \approx 1.457$): $(3 - 3(1.457), 1.457, 0) \approx (-1.370, 1.457, 0)$.

Examination 2023 — Q8(c): Ray–Sphere Intersection [3.5 marks]

Question: Let S_1 be a sphere of radius 8 centered at $(2, 4, 1)$ and S_2 a sphere of radius 10 centered at $(10, -2, -5)$. Determine if a ray with $\mathbf{s} = 2\mathbf{I} + 5\mathbf{K}$ and $\mathbf{d} = \mathbf{I} - 2\mathbf{J}$ intersects the spheres.

Note the different $\mathbf{s}, \mathbf{d}$ from 2021's Q8(e) (same sphere centers/radii, different ray). $A = |\mathbf{d}|^2 = 1 + 4 = 5$. S_1 : $\mathbf{s} - \mathbf{c}_1 = (0, -4, 4)$, $B = 0 + 8 + 0 = 8$, $C = 0 + 16 + 16 - 64 = -32$. $B^2 - AC = 64 + 160 = 224 > 0$: $t = \frac{-8 \pm \sqrt{224}}{5} \Rightarrow t_1 \approx 1.393$, $t_2 \approx -4.593$ — signs differ $\Rightarrow$ ray starts **inside** S_1 , exits at $t \approx 1.393$: point $\approx (2 + 1.393, 0 - 2(1.393), 5) = (3.393, -2.787, 5)$. S_2 : $\mathbf{s} - \mathbf{c}_2 = (-8, 2, 10)$, $B = -8 - 4 + 0 = -12$, $C = 64 + 4 + 100 - 100 = 68$. $B^2 - AC = 144 - 340 = -196 < 0 \Rightarrow$ **no intersection** with S_2 . **Conclusion:** the ray intersects only S_1 (once, since it originates inside), and misses S_2 entirely.

Examination 2024 — Q8(b), Q8(c) & Q8(d)

Q8(b) — Why Ray Tracing Is Efficient [2 marks]

Question: Why is Ray Tracing an efficient method of generating images?

Per §1.3: a single ray/object intersection test simultaneously performs **projection** (rays act as projectors), **hidden-surface removal** (only the nearest hit is kept), **shading/shadowing** (local contribution via a shadow ray), and can capture **reflection/refraction** (recursive secondary rays) — one unified computational process

instead of separate pipeline stages for each of these tasks.

Q8(c) — Ray vs Vector [2 marks]

Question: Distinguish between ray and vector.

See §1.1 (same answer as 2021 Q8c/2022 Q8a): vector = direction+magnitude, position-free; ray = direction+fixed starting point, one-sided ($t \geq 0$).

Q8(d) — Point/Plane Same-Side Test [2+2 marks]

Question: Given the plane $5x - 3y + 6z = 7$: (i) find the normal vector to the plane, and (ii) determine whether $P_1(1, 4, 2)$ and $P_2(-5, -1, 3)$ are on the same side of the plane.

(i) The plane is in the form $ax + by + cz = d$ with $(a, b, c) = (5, -3, 6)$, so the normal vector is $\mathbf{n} = 5\mathbf{I} - 3\mathbf{J} + 6\mathbf{K}$. (ii) Define $f(x, y, z) = 5x - 3y + 6z - 7$ (see §1.5):

$$f(P_1) = 5(1) - 3(4) + 6(2) - 7 = 5 - 12 + 12 - 7 = -2 \quad f(P_2) = 5(-5) - 3(-1) + 6(3) - 7 = -25 + 3 + 18 - 7 = -11$$

Both negative (same sign) $\Rightarrow P_1$ and P_2 are on the **same side** of the plane.

Quick Summary

Subtopic	Years	Method
Ray vs vector	2021, 2022, 2024	Vector = direction+magnitude; ray = direction+fixed origin, one-sided
Ray equation point-finding	2021, 2022	Plug t into $\mathbf{r}(t) = \mathbf{s} + t\mathbf{d}$ component-wise
Ray-sphere intersection	2021, 2022, 2023	Quadratic $At^2 + 2Bt + C = 0$; sign/value of roots tells enter/exit/inside/miss
Ray tracing integrates hidden-surface+projection	2021, 2022 (related), 2024	Primary ray = projector; nearest hit wins (auto hidden-surface removal)
Point/plane same-side test (3D)	2024	$f(P) = ax + by + cz - d$; same sign at two points = same side

Exam-day takeaway: every ray question reduces to one of two mechanical recipes — plug t into $\mathbf{r}(t) = \mathbf{s} + t\mathbf{d}$, or set up the quadratic $At^2 + 2Bt + C = 0$ and read off enter/exit/inside/miss from the signs of the roots. The ”why efficient / integrates hidden-surface+projection” explain-questions all boil down to the single idea in §1.3: the nearest ray/surface hit does projection and hidden-surface removal for free.